Visa Approval Facilitation

Ensemble Method & Model Tuning

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Executive Summary: Recommendations

XGboost model produced the best results

An XGboost model should be used to facilitate the decision making process of EasyVisa's visa applications

Most Important Trait of Good Applicants

Hiring highly educated people is the single most differentiable factor in an applications

Correlations are symptomatic but not necessarily causal

Many of the factors can not be reduced to the most significant factor because they are interconnected in and of

Business Problem Overview

775,979 Applications

were submitted by employers seeking labor certifications.

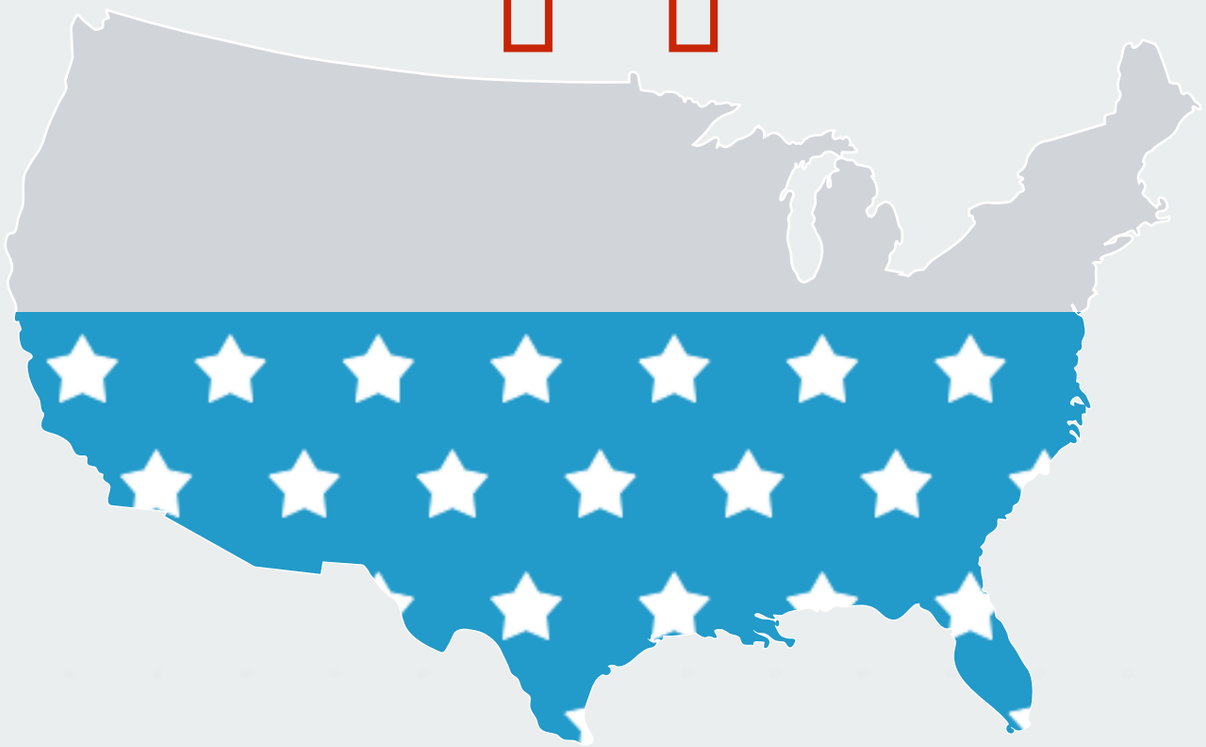
EasyVisa has been hired!

The Company's Goal:

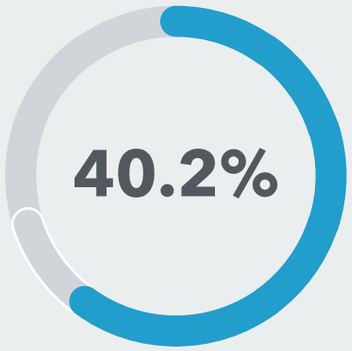
1. Facilitate the process of visa approvals.
2. Recommend a suitable profile for the applicants whom the visa should be certified or denied to based on the drivers that significantly influence the case status.

Exploratory Data Analysis

66.8% of Visas were approved



Visa Approval Rate by Education



Had a Bachelor's

According To The World Bank, this is much higher than the average in most countries - including the US..

Had a High School Diploma

A low percentage - suggesting that most employers are seeking to bring intellectual labor to the US

Had a Master's

However, the meaning of this data is hard to interpret because their is no universal accreditation system just as there is no universal church

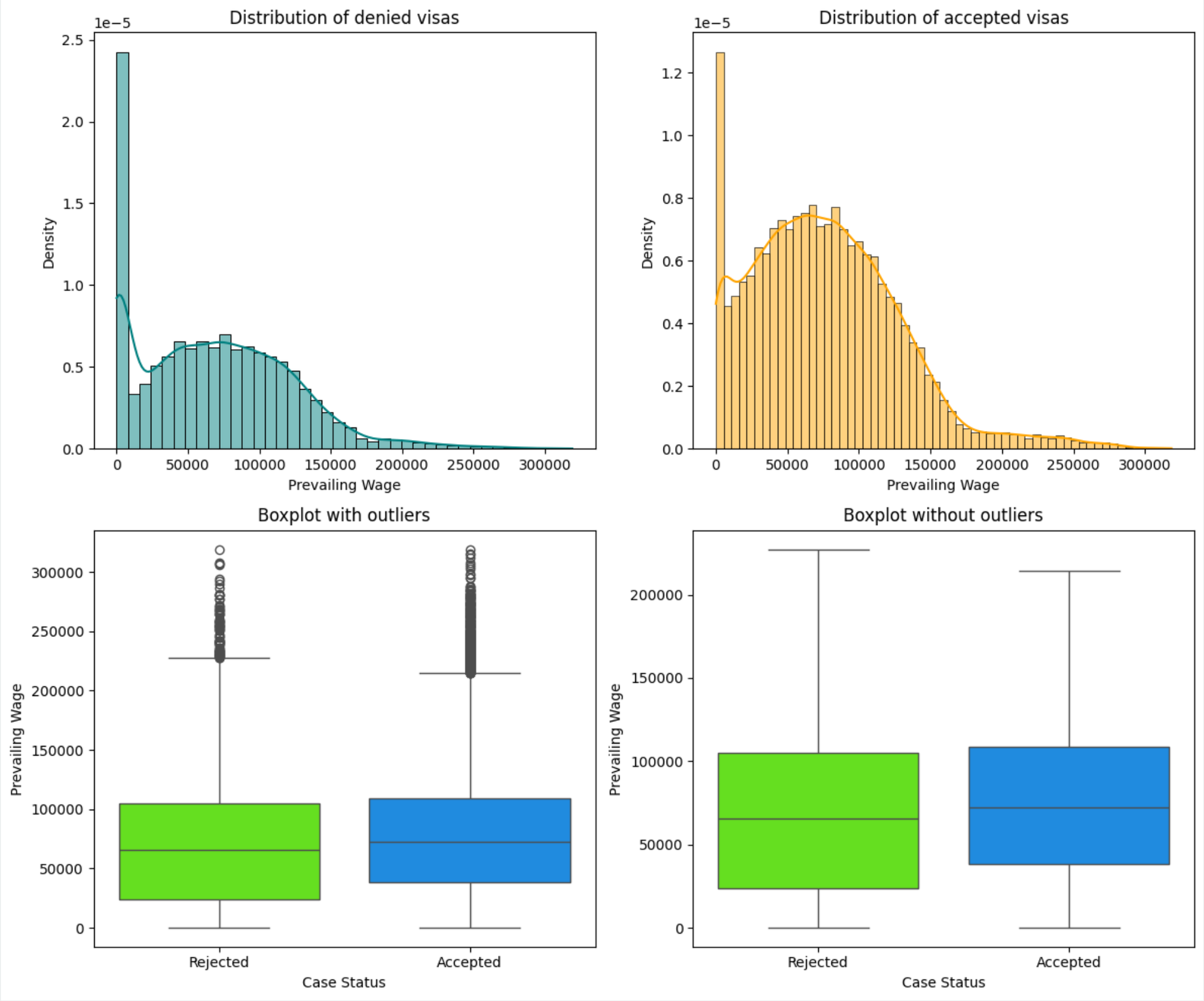
Had a Doctorate

Which is not too surprising given the cost of acquiring a doctorate and the cost of moving abroad

SOURCE: WORLD BANK GROUP. HTTPS://DATA.WORLDBANK.ORG/INDICATOR/SE.TER.CUAT.BA.ZS.

Visa by Wage

The difference between who is accepted and rejected does not reveal much of a difference in wages



US Visa Approval by Region

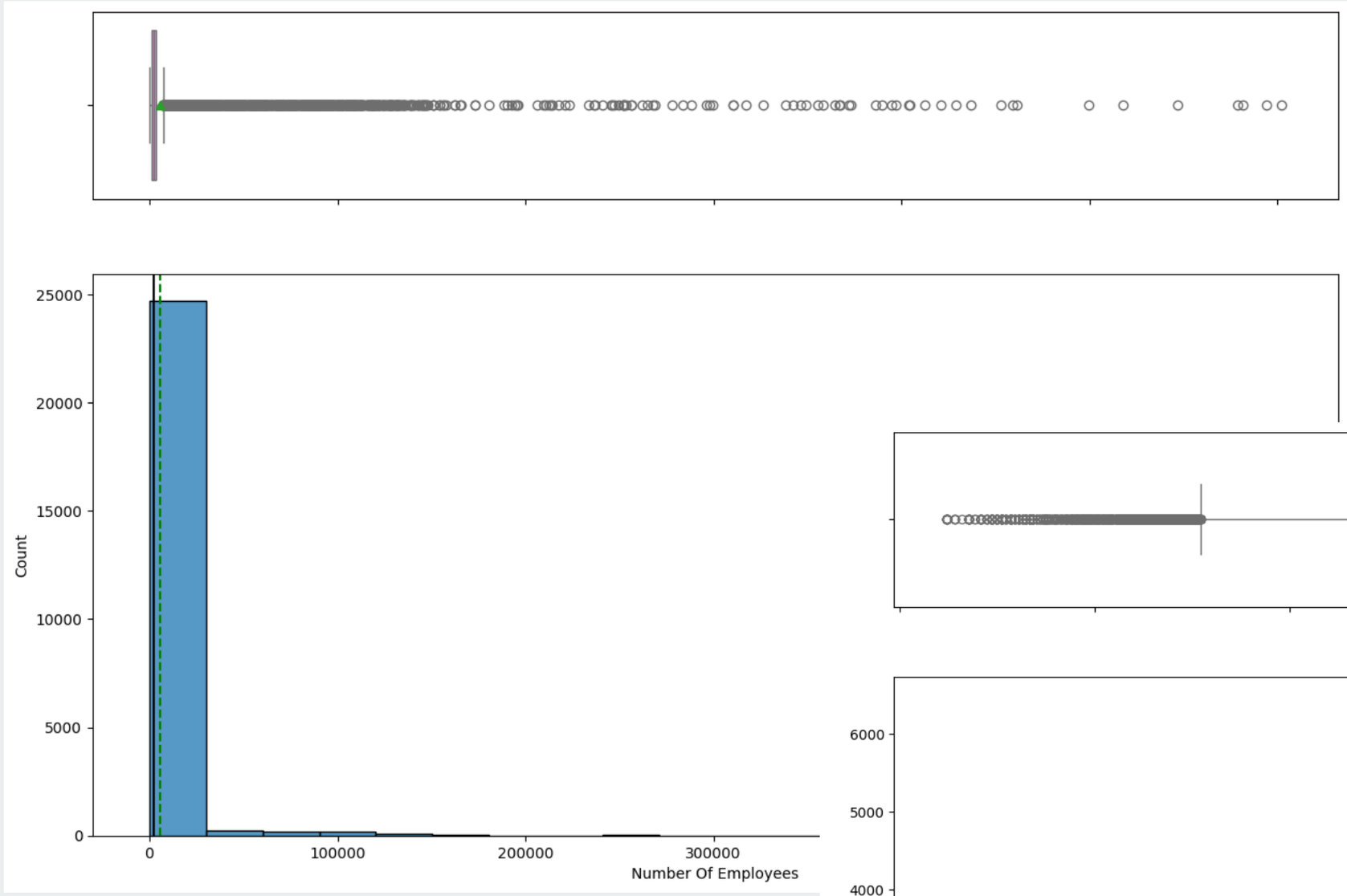
16.9% in the Midwest

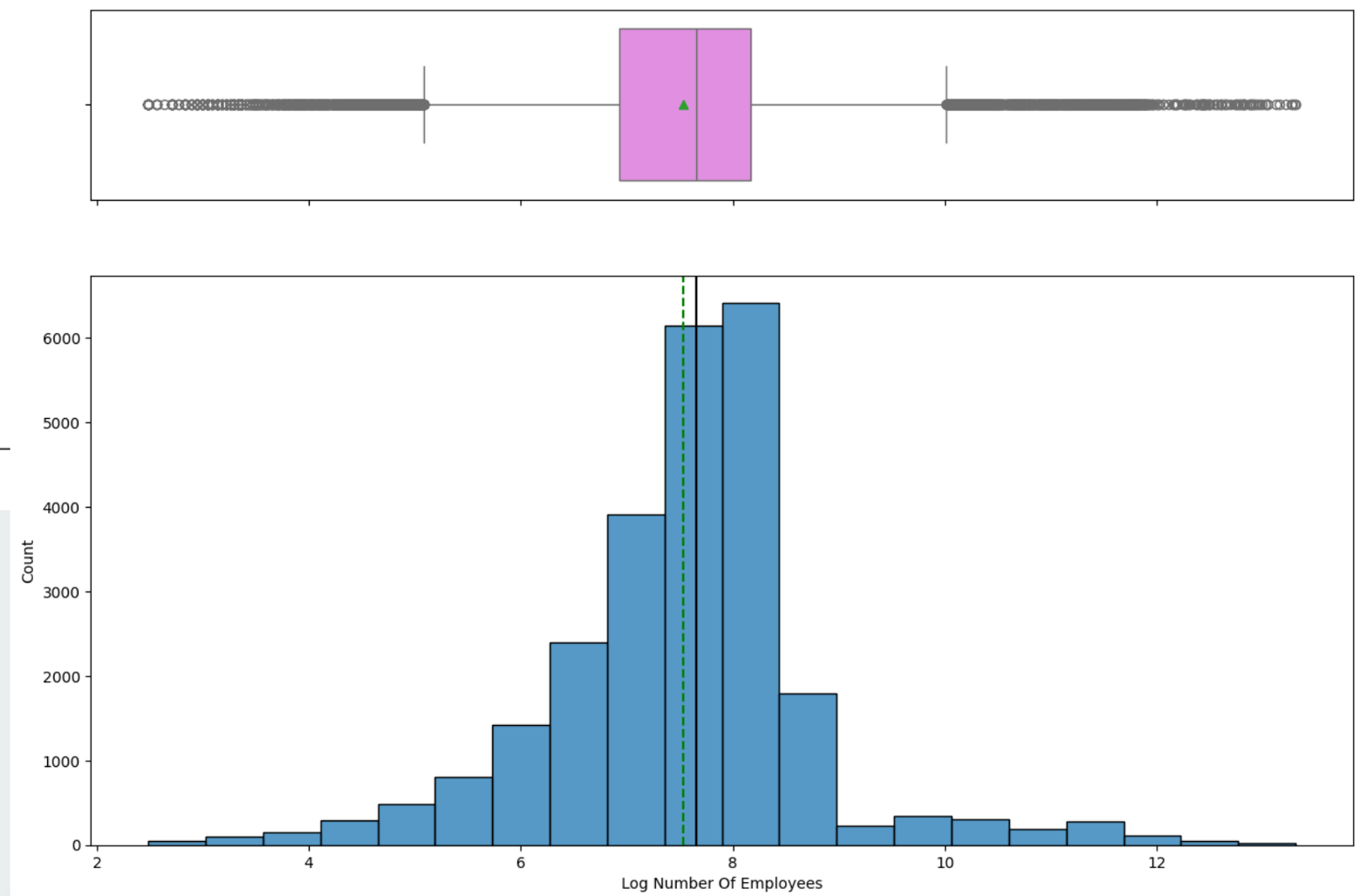
28.2% in the Northeast

25.5% out West

27.5% in the South

& 1.5% live in Hawaii

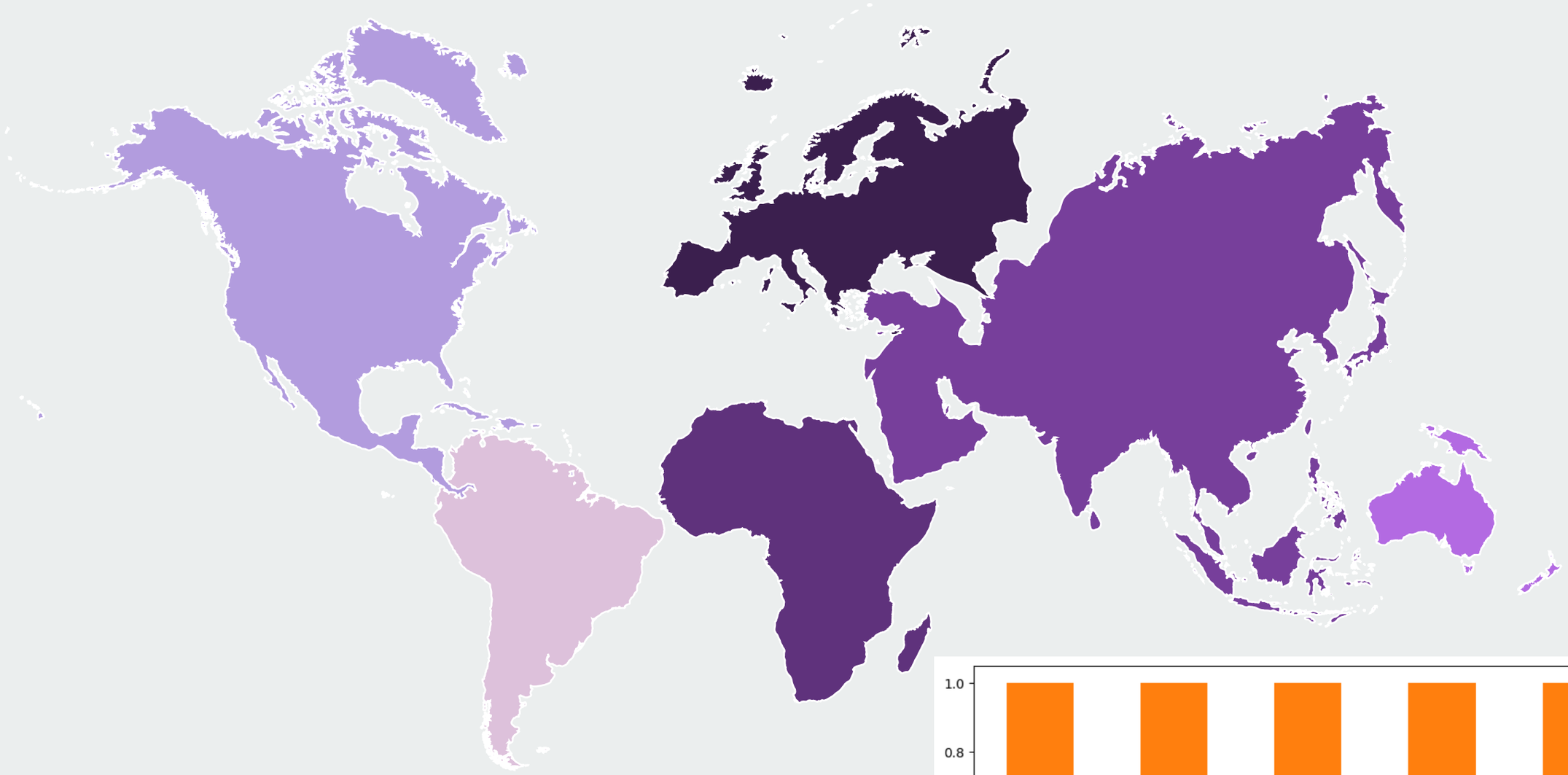


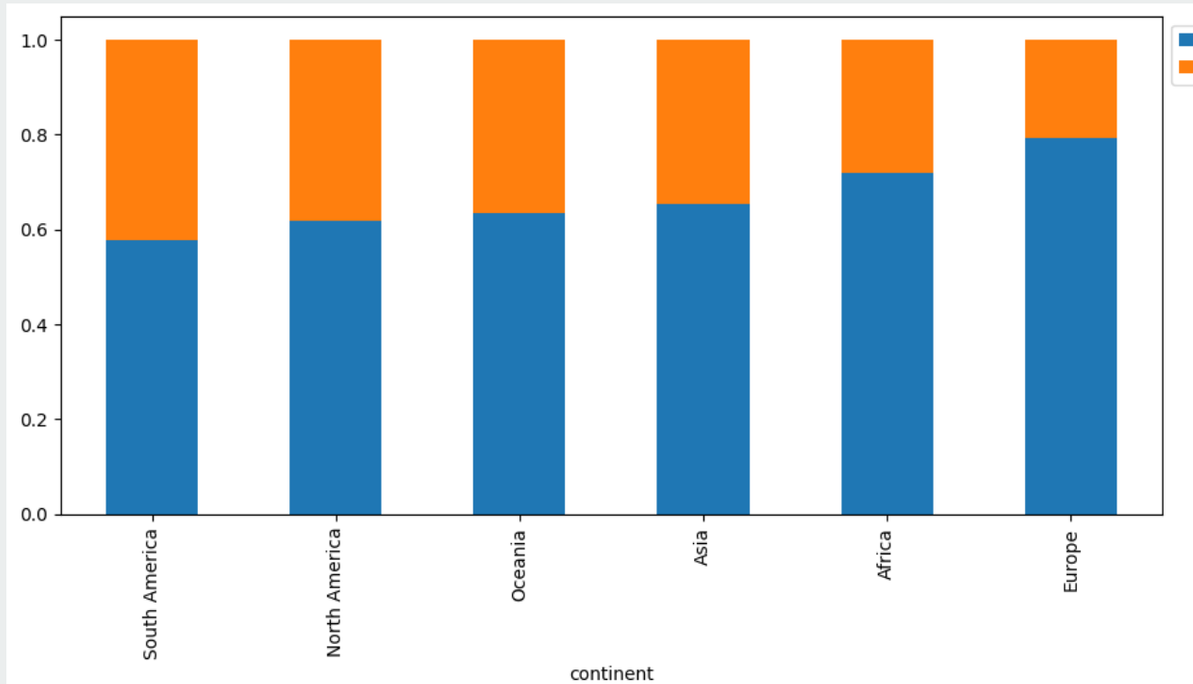


Visa Approval by Number of Employees

Most Applications are for fewer than 250k employees.
The chart on the right is more informative

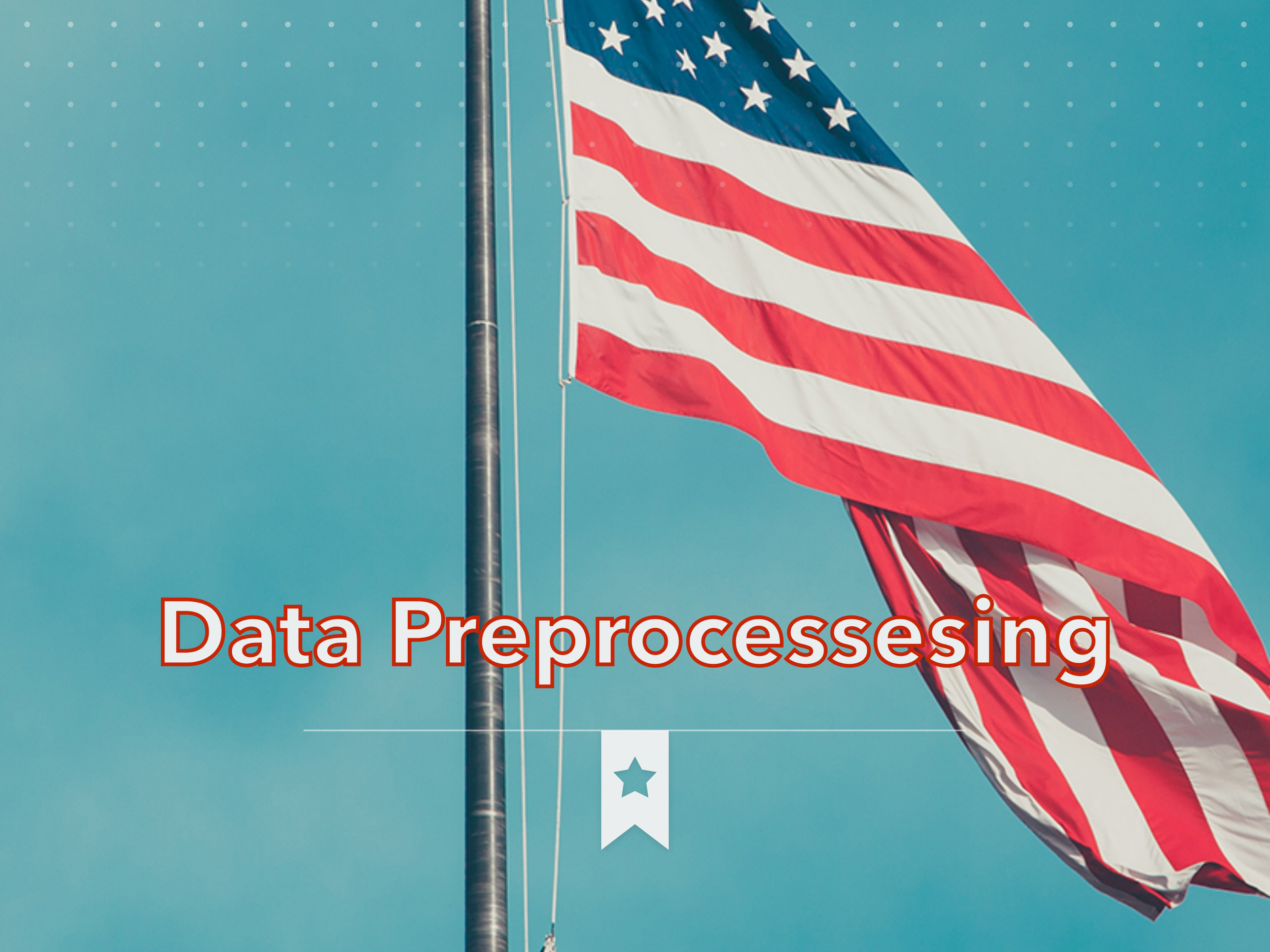
Visa Approval by Continent



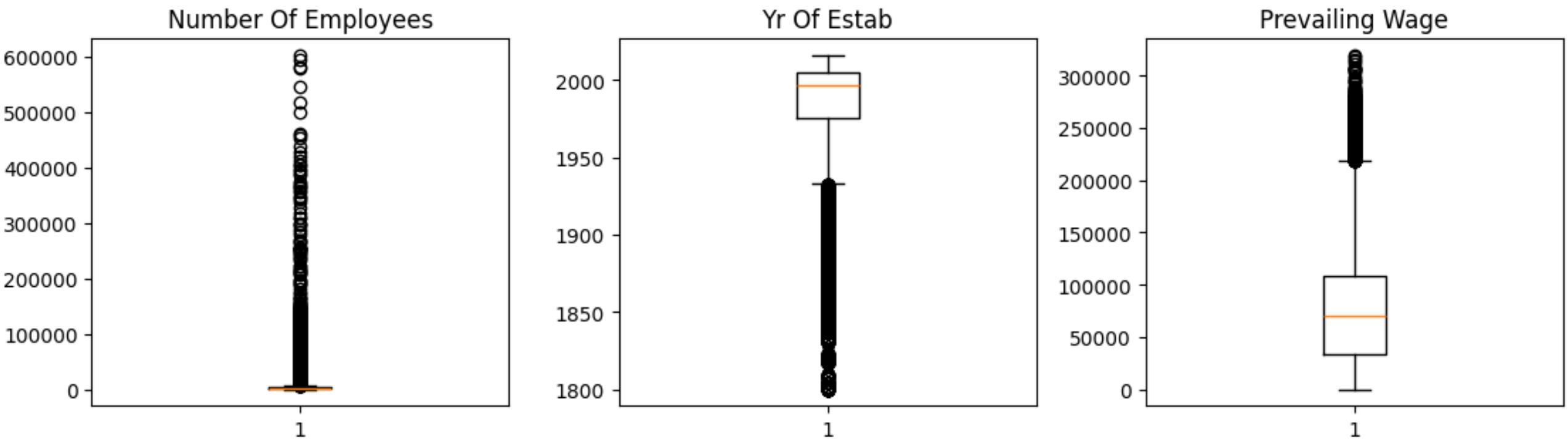


There is not much difference in acceptance rate by continent

However, companies from Europe and Africa are the most likely to have their visa applications accepted



Data Preprocessesing



Missing Values & Outlier Treatment

None of the outliers that appear here seem extremely out of the ordinary, and they seem like they contain valid data points.

Moreover, there were no null nor duplicated values.

Feature Engineering

DATA CLEANING

SPLIT THE DATA

Into a training, validation, and test set

CHECK THE SIZE OF EACH SET

Each set has about 67% accepted and 33% rejected applicants

REASSESS GOAL

The goal is to improve both the precision and recall - as is represented by the F1 score

Data Prep without Leakage

REMEMBER

No outlier can be removed or treated now that the data has been split - in order to avoid data

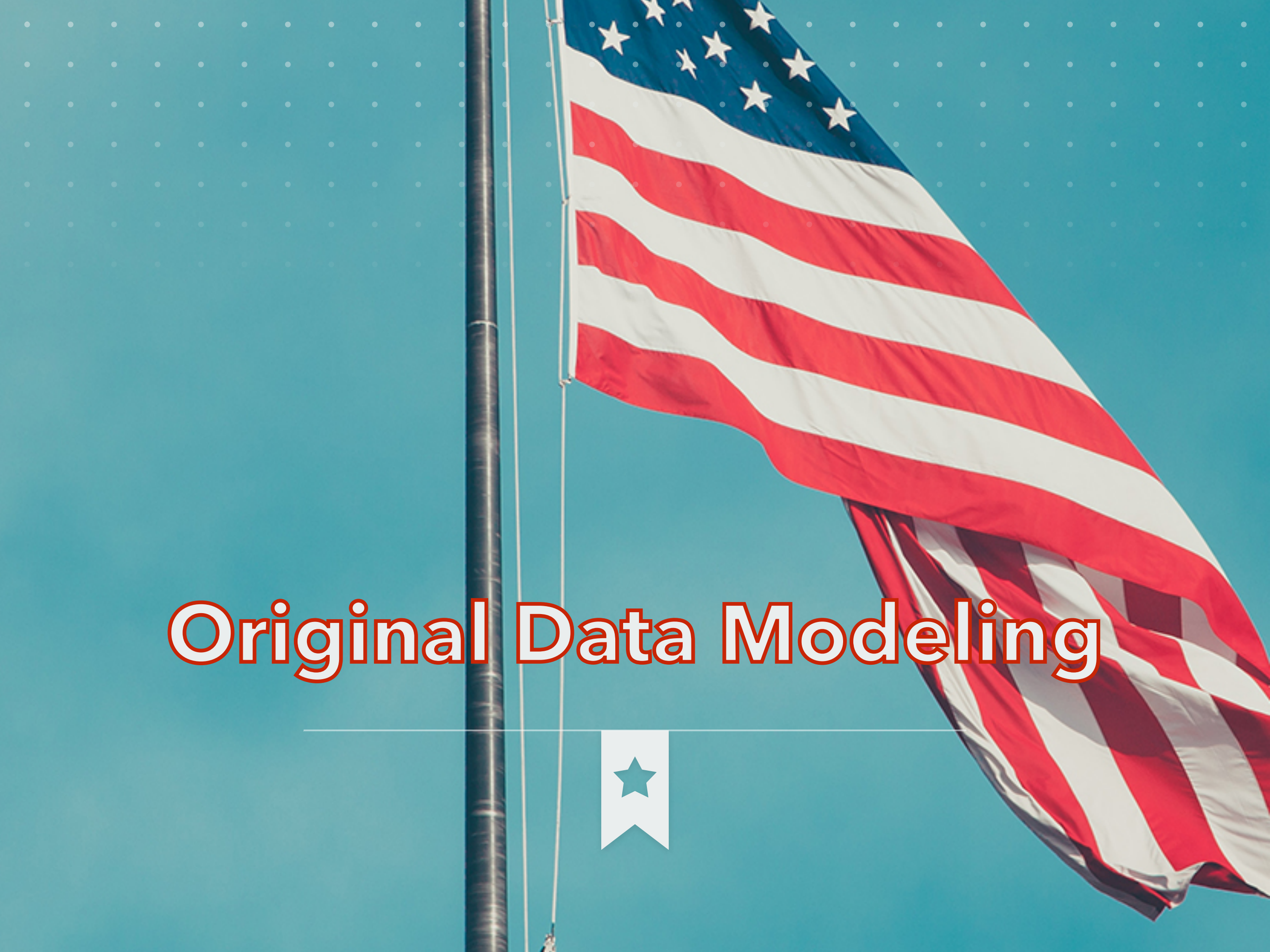
CONSTRUCT MODELS

Using bagging, random forest, adaboost, gradient boosting, and XGboost

REPEAT

Once for the original data, then by oversampling and by under sampling

EVALUATE MODELS



Original Data Modeling

The Models

F1 scores from models made with the original data

.83 USING GRADIENT BOOSTING

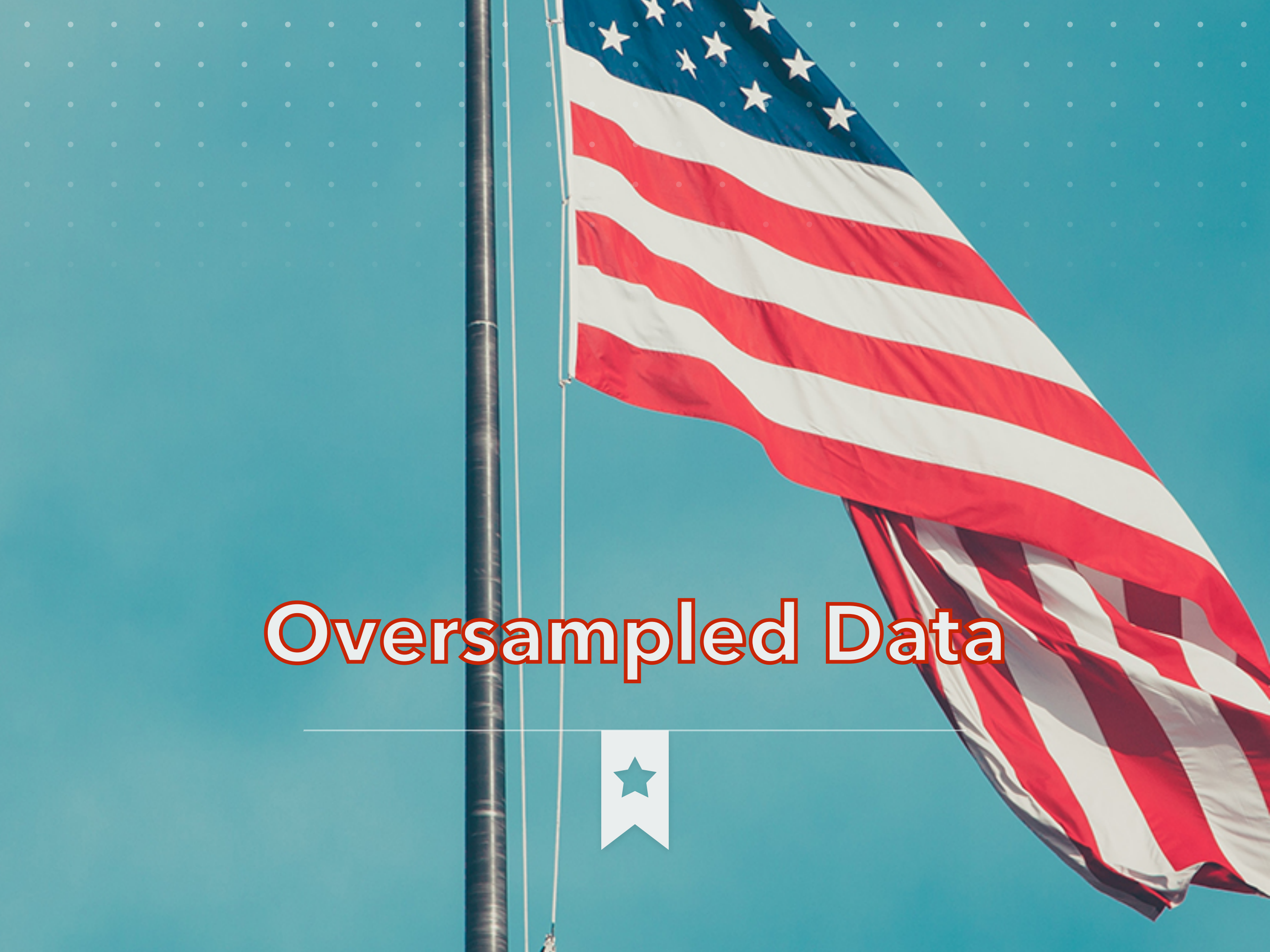
.82 USING ADABOOST

.81 USING XGBOOST

.80 USING RANDOM FOREST

.77 USING BAGGING

All of the models performed quite similarly; however, Gradient Boosting produced the best results



Oversampled Data

The models produced nearly identical F1 scores relative to the original data sampling

.82 USING GRADIENT BOOSTING

.82 USING ADABOOST

.81 USING XGBOOST

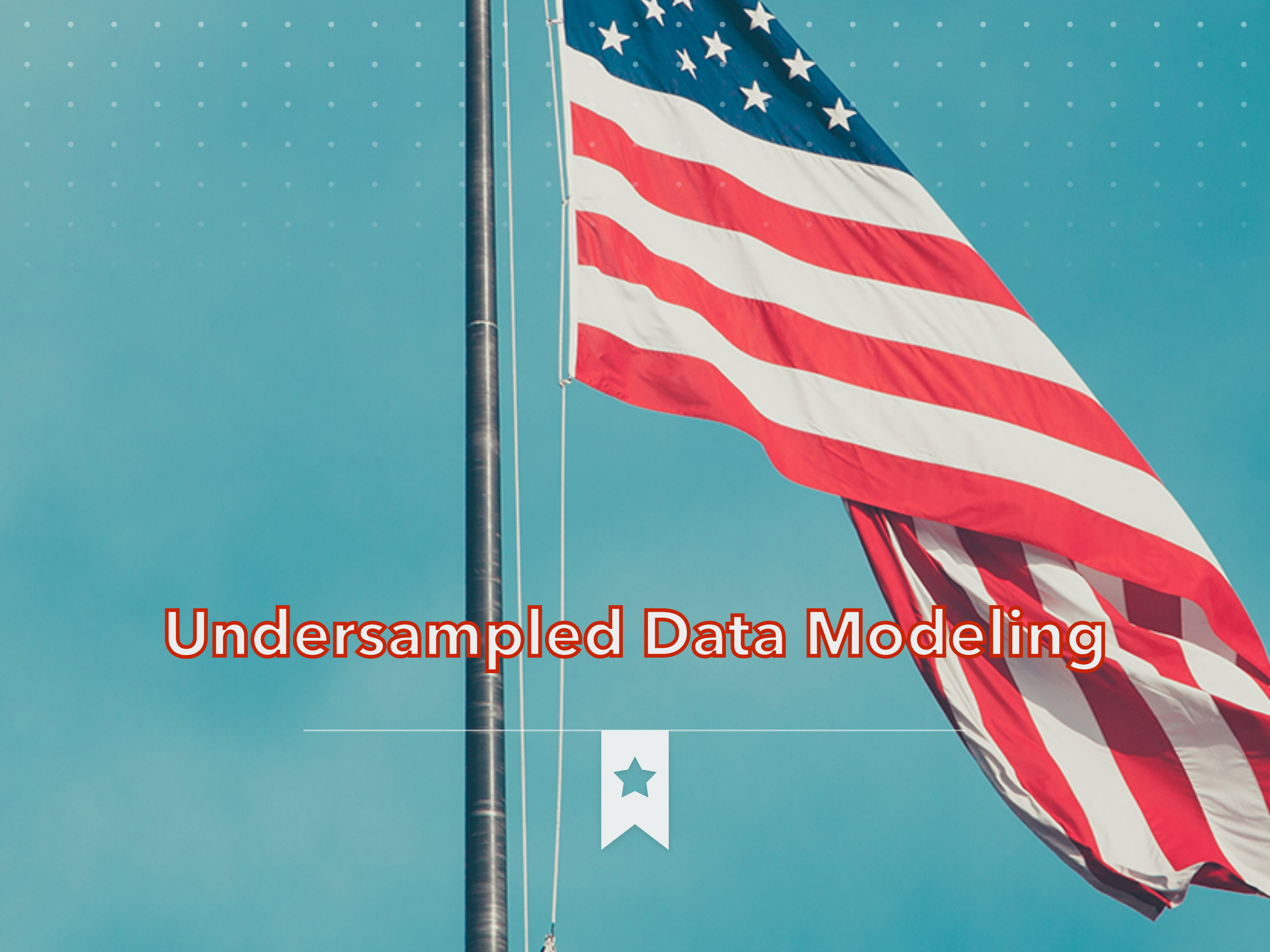
.80 USING RANDOM FOREST

.77 USING BAGGING

All of the models performed quite similarly; however, AdaBoost just barely produced the best results

The Models

F1 scores from models made with the oversampled data



Undersampled Data Modeling

The Undersampled Data produced worse results

The Models

F1 scores from models made with the undersampled data

.78 USING GRADIENT BOOSTING

.77 USING ADABOOST

.75 USING XGBOOST

.74 USING RANDOM FOREST

.71 USING BAGGING

All of the models performed quite similarly; however, Gradient Boosting produced the best results

Hyperparameter Tuning

Models To Be Tuned

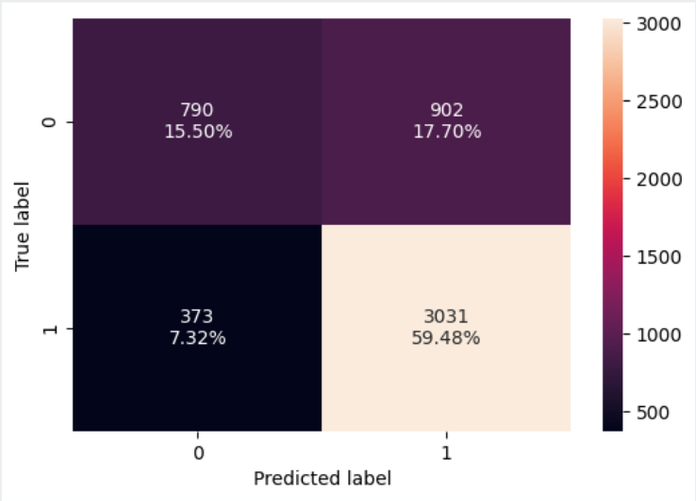
I choose 3 models that I thought could be improved with some hyper parameter tuning:

Original Data Set's Random Forest Model

Original Data Set's AdaBoost Model

Original Data Set's XGboost Model

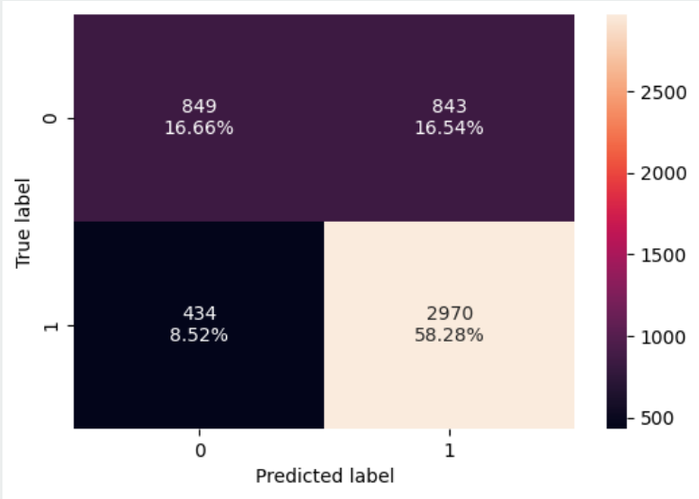
Model Comparison



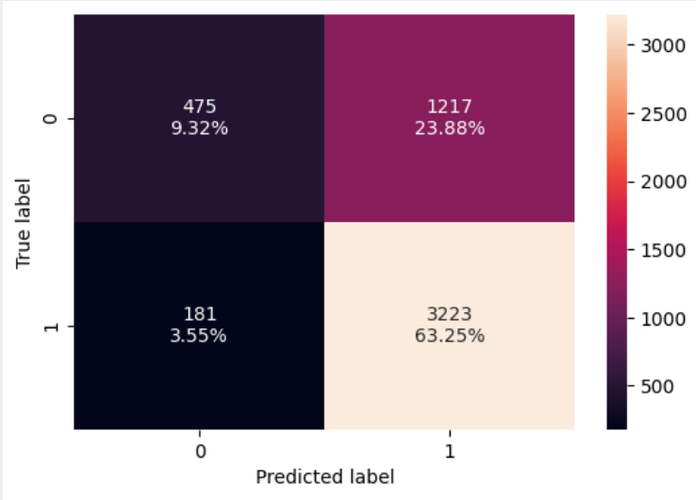
Accuracy	Recall	Precision	F1
0.749804	0.890423	0.770659	0.826223

Random Forest

Accuracy	Recall	Precision	F1
0.749411	0.872503	0.778914	0.823057



AdaBoost



Accuracy	Recall	Precision	F1
0.725667	0.946827	0.725901	0.821775

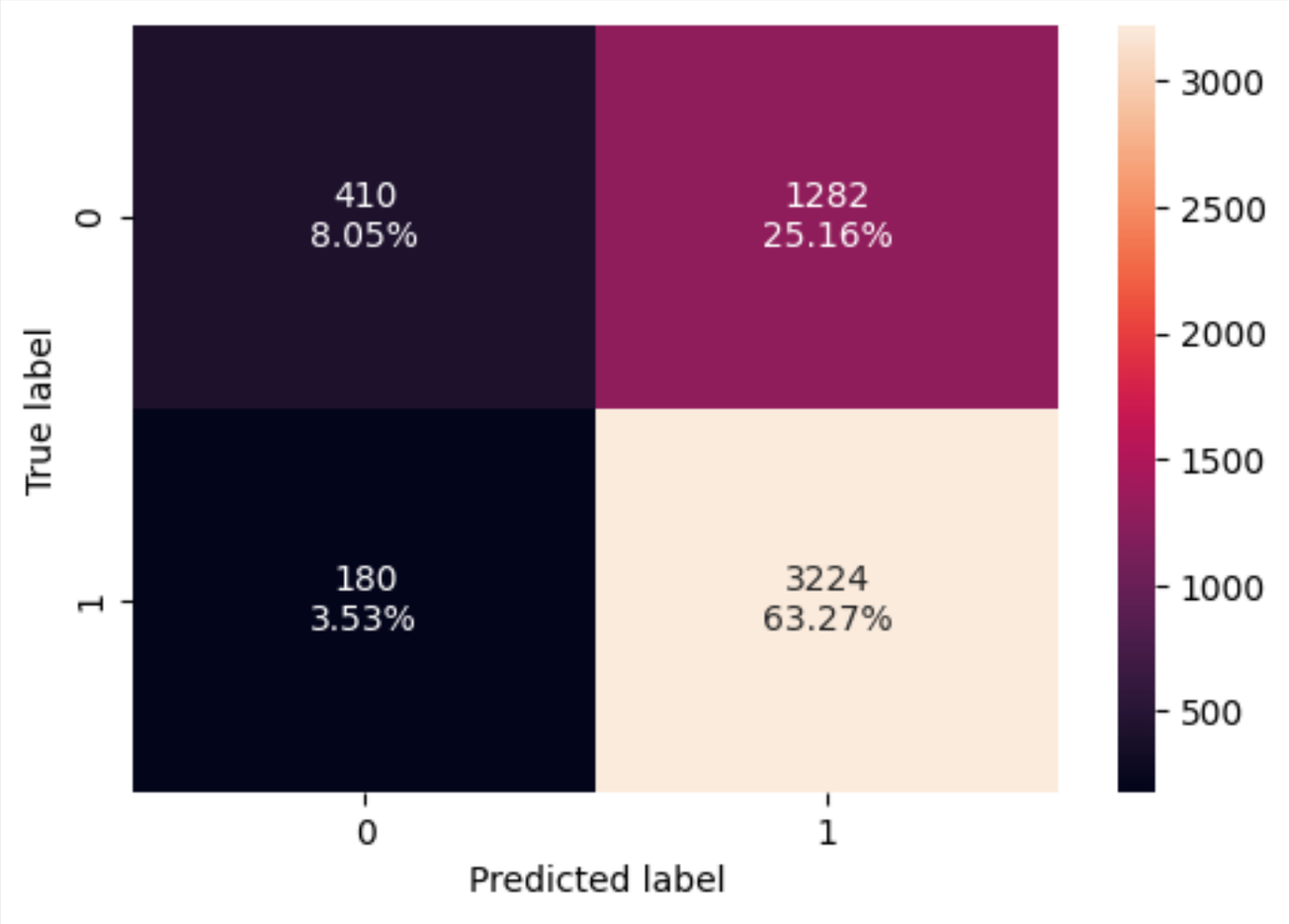
XGboost

The Best Model:

The Original Data Set's XGboost model

The Performance of The Best Model

Here is how the model performed with the test set data!

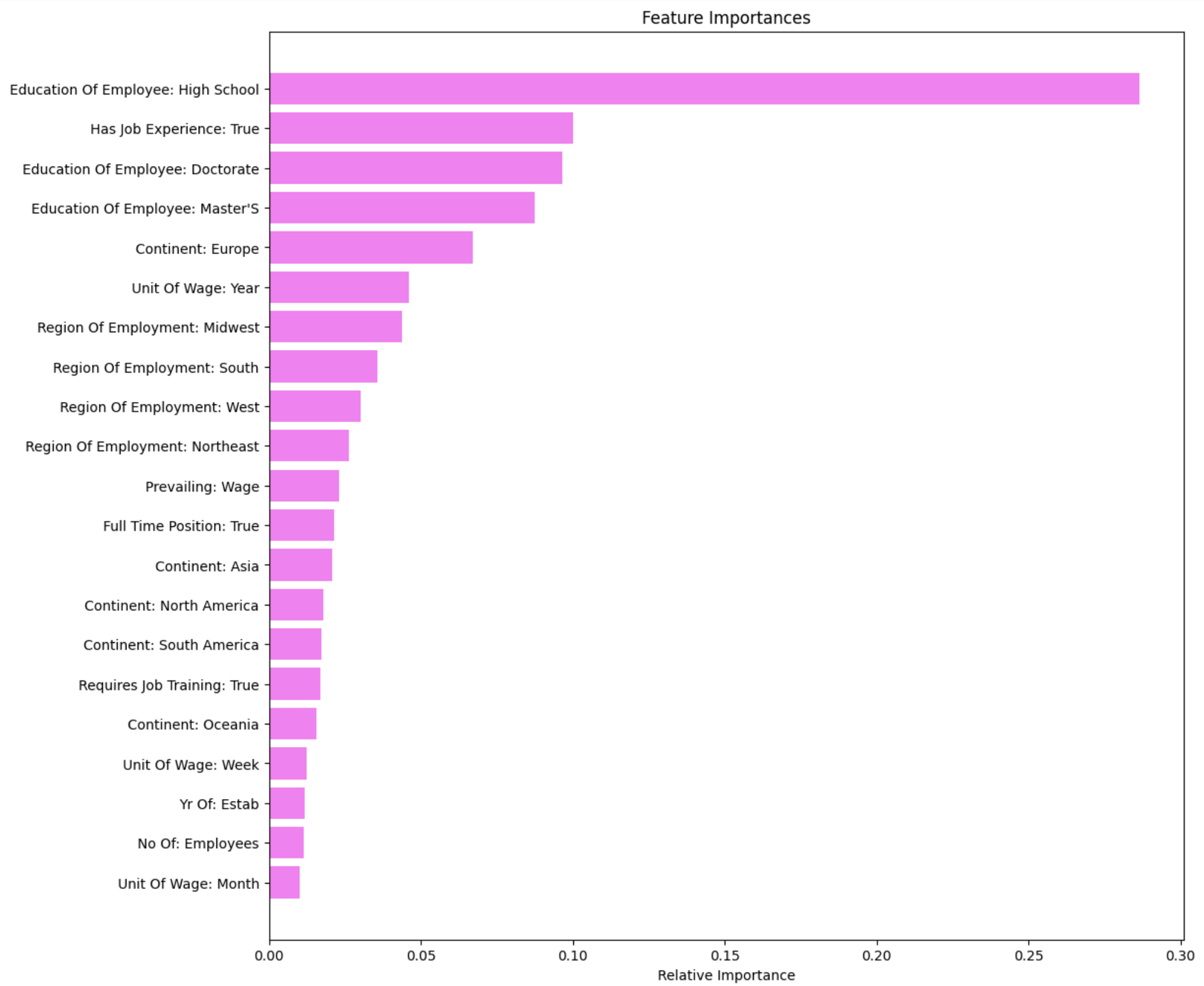


Accuracy	Recall	Precision	F1
0.713108	0.947121	0.71549	0.815171

Insights From The Best Model

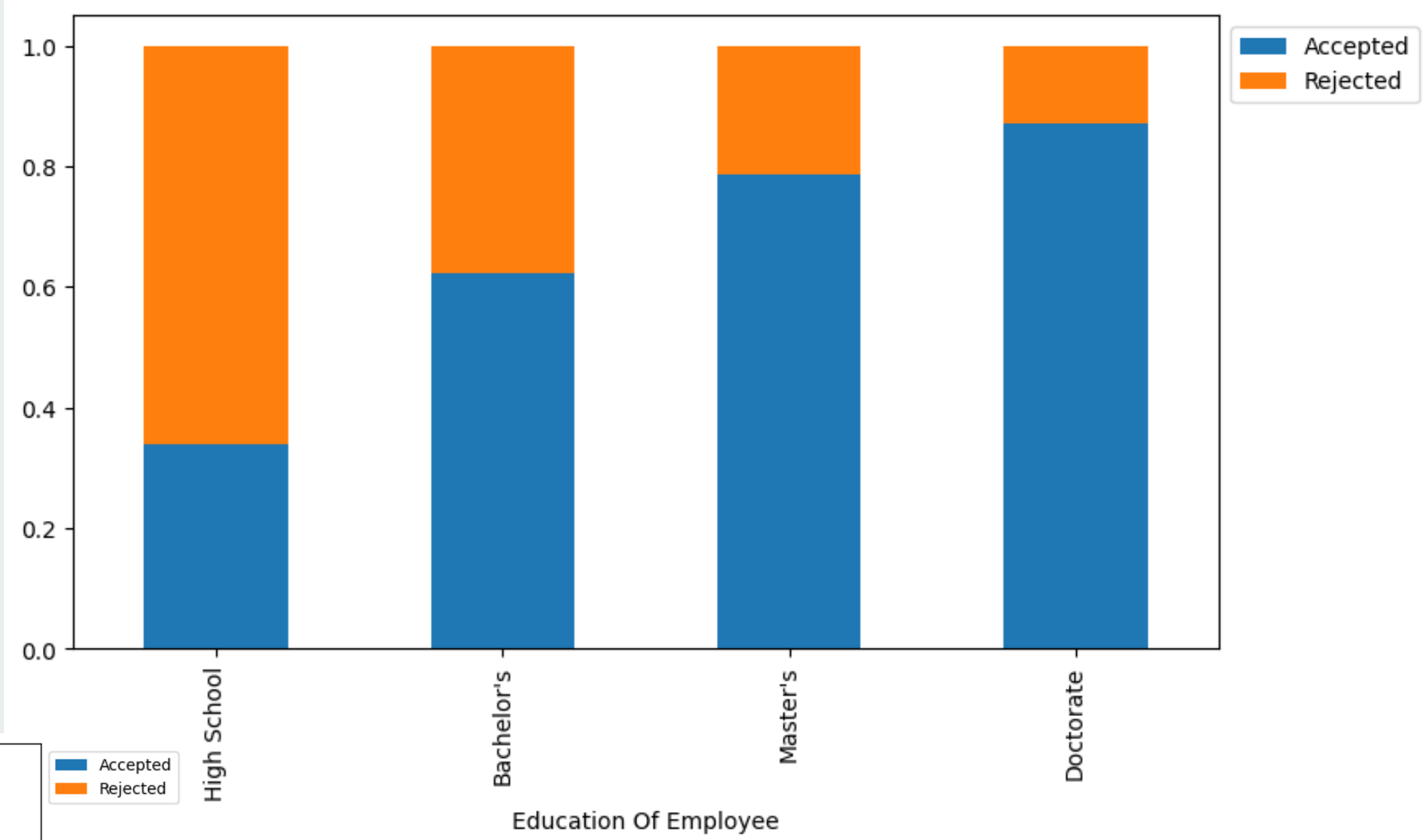
XGboost models are an ensemble of decision tress, and to the left represents the weight that each factor has on the decision trees which form its behavior.

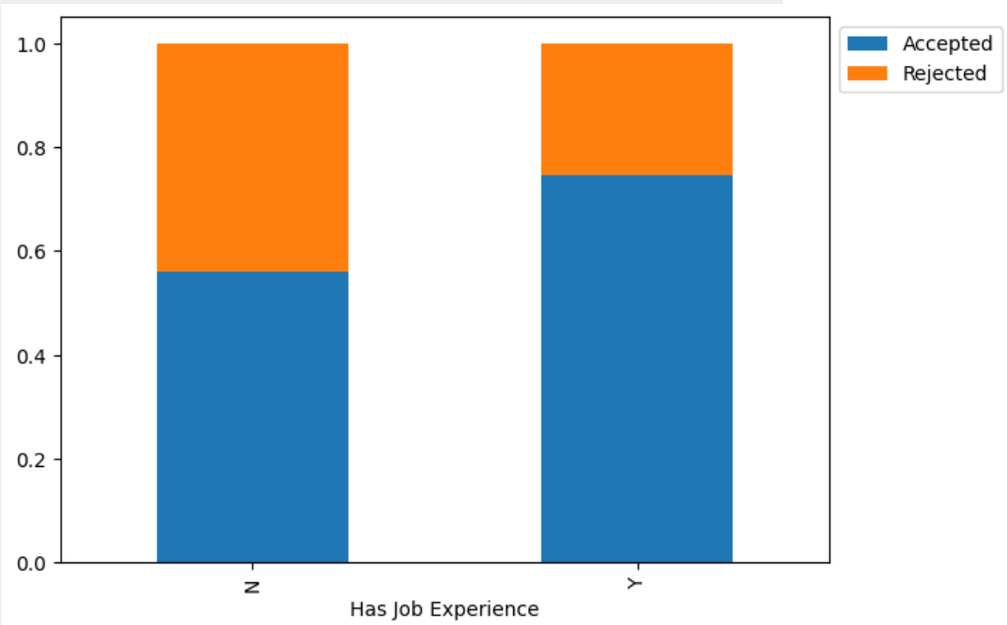
Moreover, while Europeans had a higher chance of being accepted, this might be best explained by how 22.67% of those companies were hiring people with Doctorates



Business Recommendations

The single most differentiable thing to consider when deciding whether to accept an application is whether or not the employer is hiring people with more education





There should be close consideration of whether the employer is hiring people with job experience when reviewing visa applications

Final Note: Regarding Wage

While wage wasn't ranked the most significant factor, the range of wages is great as opposed to having or not having a degree.

Wages varied from $2.14 to $31,9210.27

However, calculating for salary revealed a weak correlation between approval and salary: -0.085, with an R squared value of 0.0072.

Thank you!